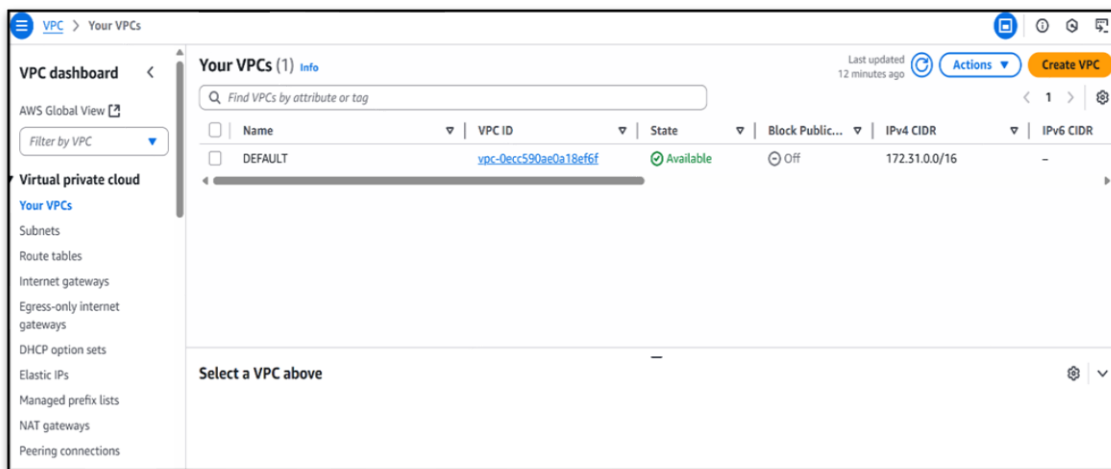


ASSIGNMENT 1

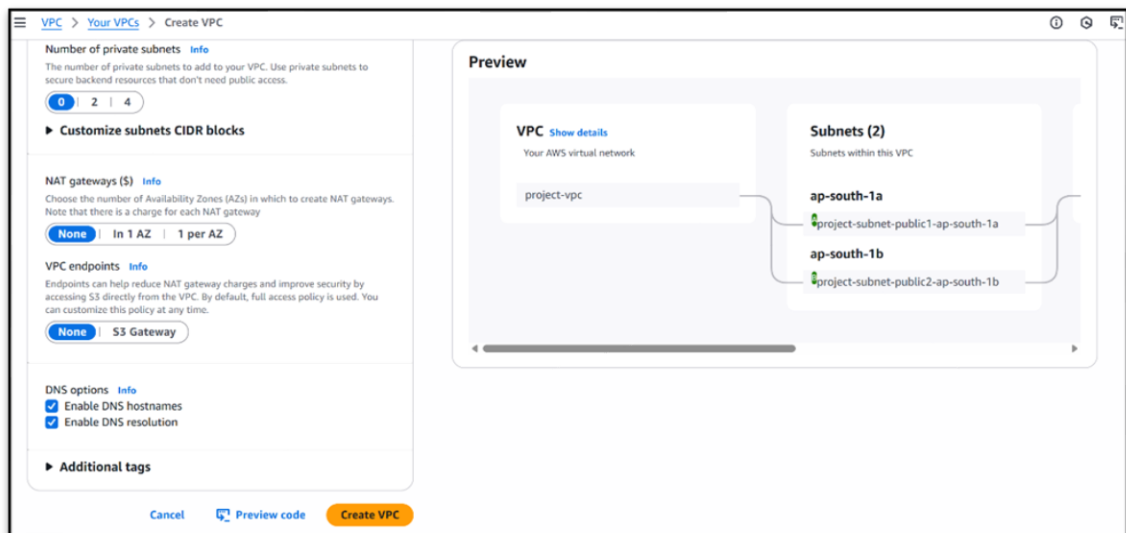
Task 1:

To create VPC with 2 subnets and launch 2 instances to check the communication between two Instances:

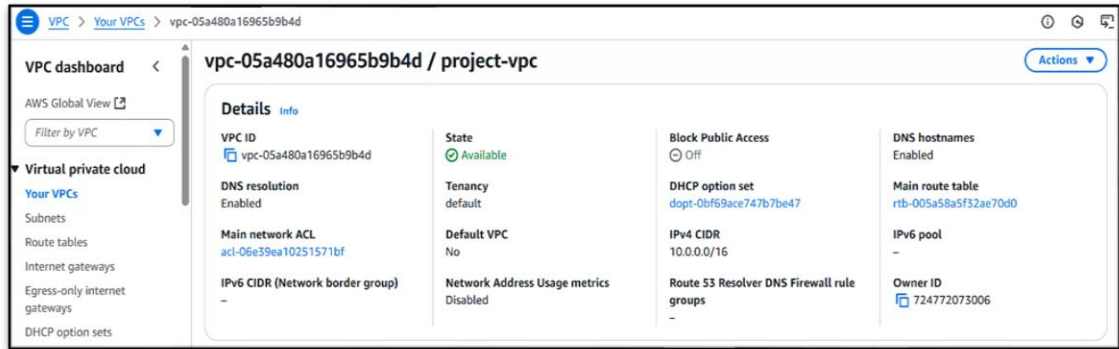
- Select VPC → Your VPC → Create VPC → Select VPC and more as it would make the process simpler.



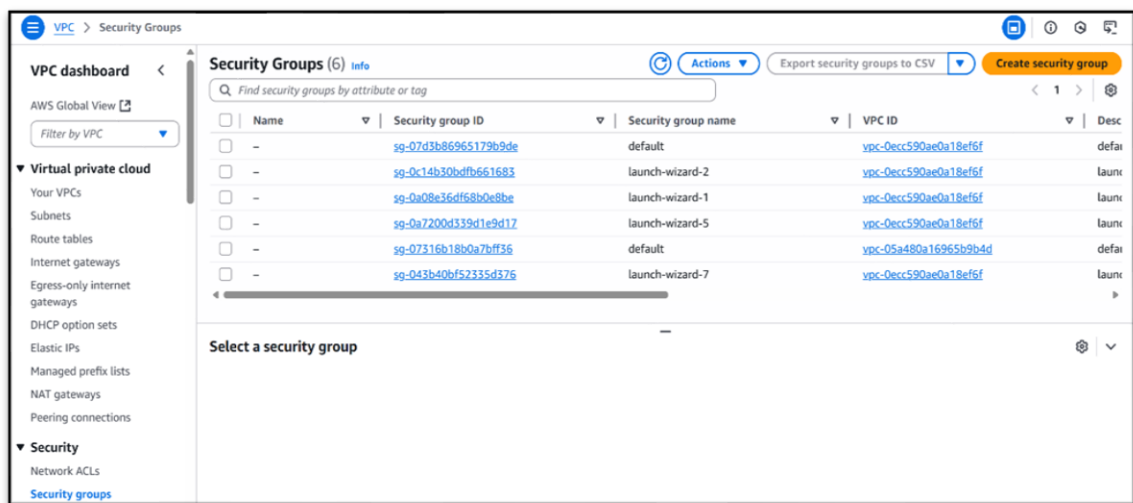
- Auto generate or give a name to the VPC → Enter the IP address → Select the No of. availability zone as 2 as it helps in redundancy.



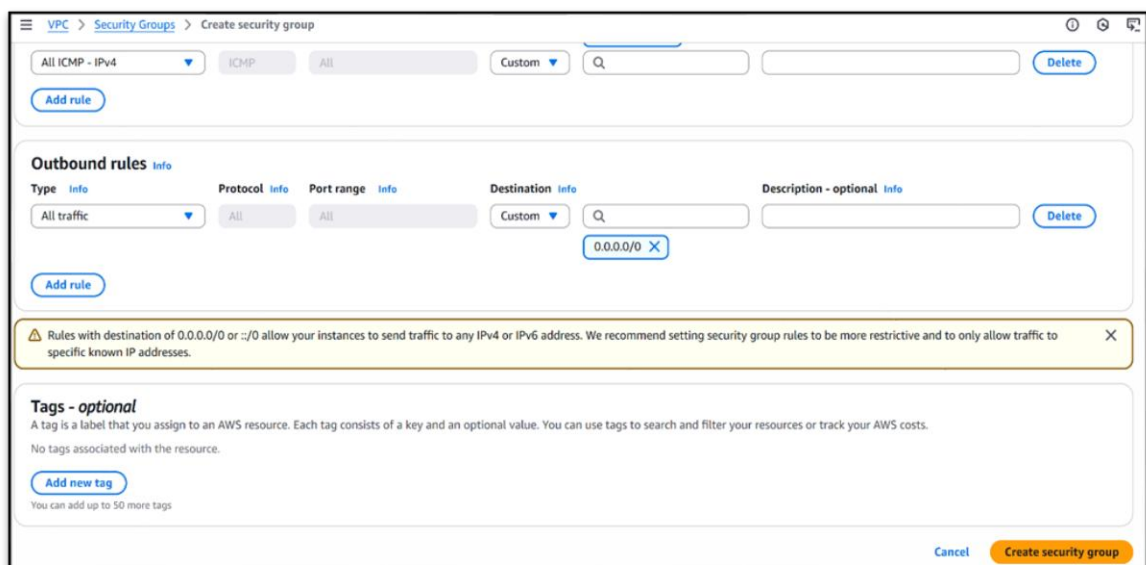
- Select two public subnets and choose all the other options as none as it is not necessary for this scenario → Create VPC.

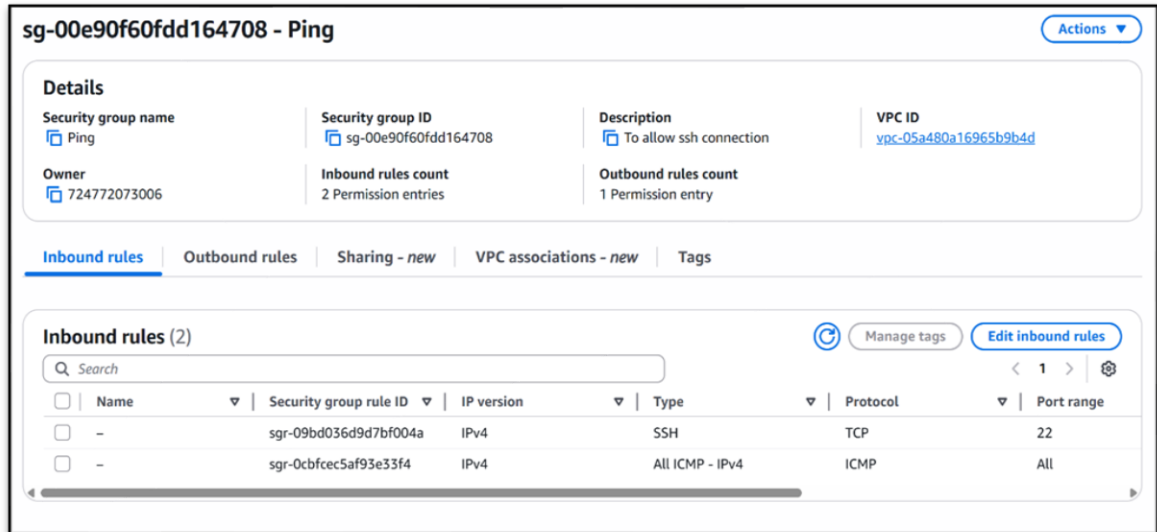


- The VPC has been successfully created with two subnets and a route table.
- Next step is to create a security group to manage the traffic: Select security groups → Create security groups.



- Enter the name for It → description if needed → Select the VPC → Add inbound rules and allow ssh connection by mentioning the Source block as 10.0.0.0/16 → Create Security group.



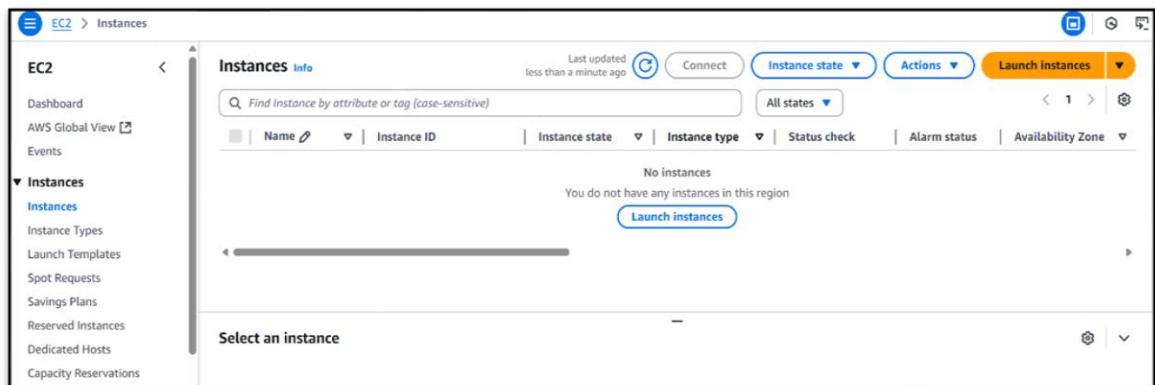


- The security group has been successfully created.

The VPC and subnet creation has been successful.

Creation of instances:

- Go to the EC2 dashboard in the console and click Launch instance.



- Enter the desired name for the instance → Select Amazon AMI → Choose instance type as t2 micro as it is free tier eligible → Select the Key pair → Edit network settings → Select your VPC → Select subnet 1 → Enable auto assign IP → Select the existing security group which we have created → Launch instance.

Network settings

VPC - required: vpc-05a480a16965b9b4d (project-vpc)

Subnet: subnet-0d1482f52dd4466a (project-subnet-public1-ap-south-1a)

Auto-assign public IP: Enable

Firewall (security groups): Select existing security group (Ping sg-00e90f60dd164708)

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023.8.2...read more

Virtual server type (instance type): t2.micro

Firewall (security group): Ping

Storage (volumes): 1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available)

Buttons: Cancel, Launch instance, Preview code

- Create another instance following the same steps above except selecting the subnet as the 2nd subnet (Not the previously selected one) → Launch instance.

Instances (2/2)

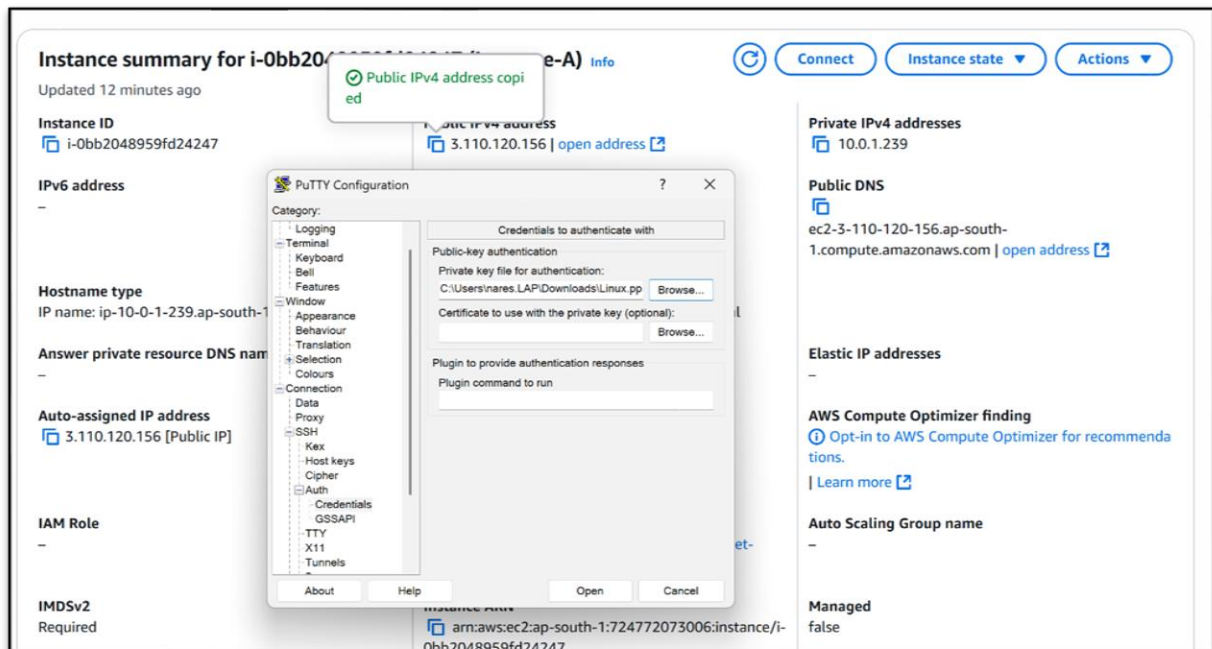
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
Instance-B	i-008c79eb67e174105	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b
Instance-A	i-0710fafc264391e16	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a

2 instances selected

- The instances have been successfully created.

To check the connection between the instances follow these steps:

- Open Putty or any other connection → In the host name enter the Public IP of instance1 → Navigate to Connection → SSH → Auth → Credentials → Browse you key pair → Open.



- After the terminal opens enter the user name as ec2-user → sudo su (super user) → ping <target IP> .

```
root@ip-10-0-1-239/home/ec2-user
login as: ec2-user
Authenticating with public key "Linux"

#
~\### Amazon Linux 2023
~~_#####\
~~_\####|
~~_\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~' '->
~~~
~~~-.-./
~/m/' - /
```

Last login: Mon Sep 15 11:05:19 2025 from 171.48.97.111
[ec2-user@ip-10-0-1-239 ~]\$ sudo su
[root@ip-10-0-1-239 ec2-user]# ping 10.0.2.222
PING 10.0.2.222 (10.0.2.222) 56(84) bytes of data.

- The connection between the two instances has been successful.